FAIRNESS AND BIAS MITIGATION

Fairness Principle

AI systems should be designed to treat all people equitably.

This involves both

- **Preventing bias** (ensuring training data doesn't contain historical prejudices reg arding race, gender, or age) and
- **inclusivity** (creating tools that work for diverse populations, not just a "default" user group).

Data bias is the systematic error introduced into data workflows and machine learning (ML) models due to inaccurate, missing, or incorrect data points which **fail to accurately represent the population**.

Example

LinkedIn's AI-driven job recommendation systems faced allegations of perpetuating gender biases. A 2022 study introduced a fairness metric to detect algorithmic bias, revealing that LinkedIn's algorithms favoured male candidates over equally qualified female counterparts, leading to unequal job recommendations.

Why:

- Tech industry is male dominated
- Even after removing gender, the AI replicates the bias because it mirrors an existing hiring bias in industry

Key Takeaway: AI agents do not "invent" new biases; they **automate and scale** existing human ones.

Common risks that undermine **fairness, Non-discrimination, and inclusion**

Unequal Errors

Looks fine overall, but fails more for some groups.

Example: the tool misses urgent cases more often for one group.

Proxy Risks

Inputs can quietly track identity.

Example: neighbourhood or school can reflect segregation.



Access and Recourse Failures

No human path. Confusing notices.

Example: appeal is online-only, so some people cannot use it.

TYPES OF BIAS IN AI

SAMPLING BIAS



ALGORITHMIC BIAS



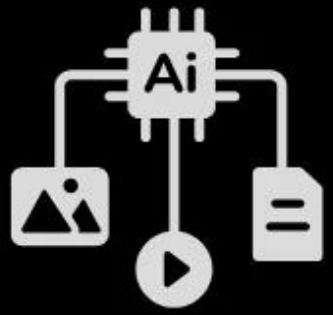
CONFIRMATION BIAS



MEASUREMENT BIAS



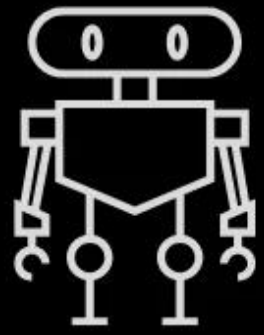
GENERATIVE BIAS



REPORTING BIAS



AUTOMATION BIAS

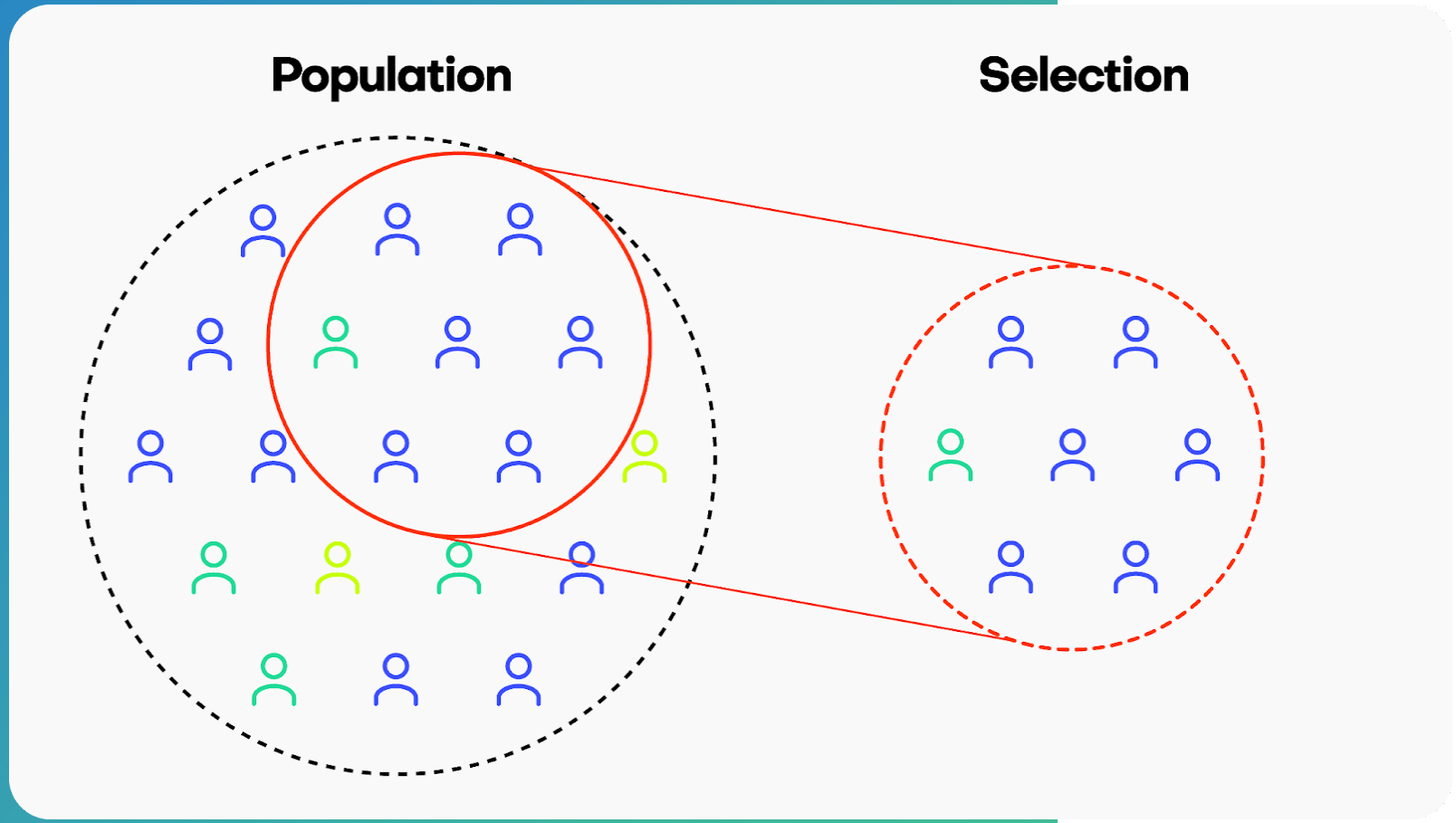


GROUP BIAS



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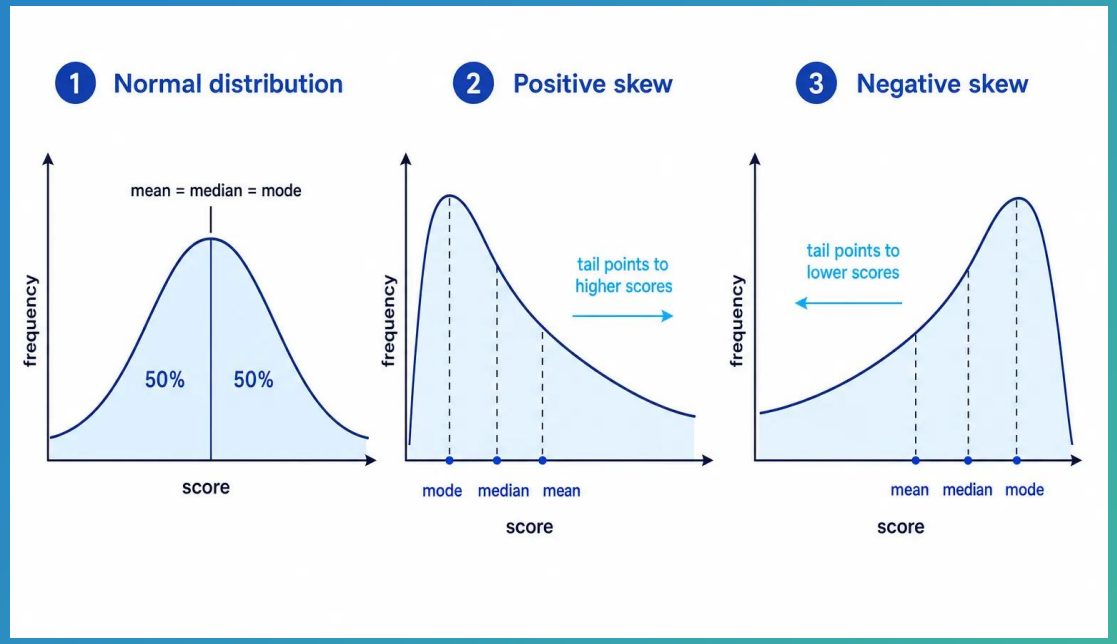
Types of data bias

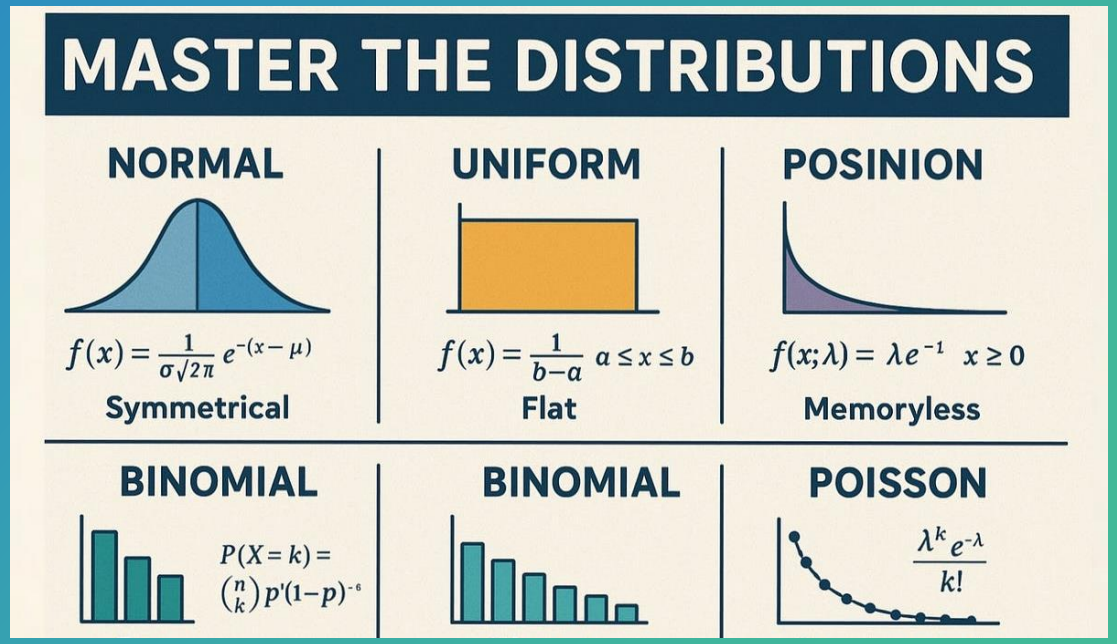


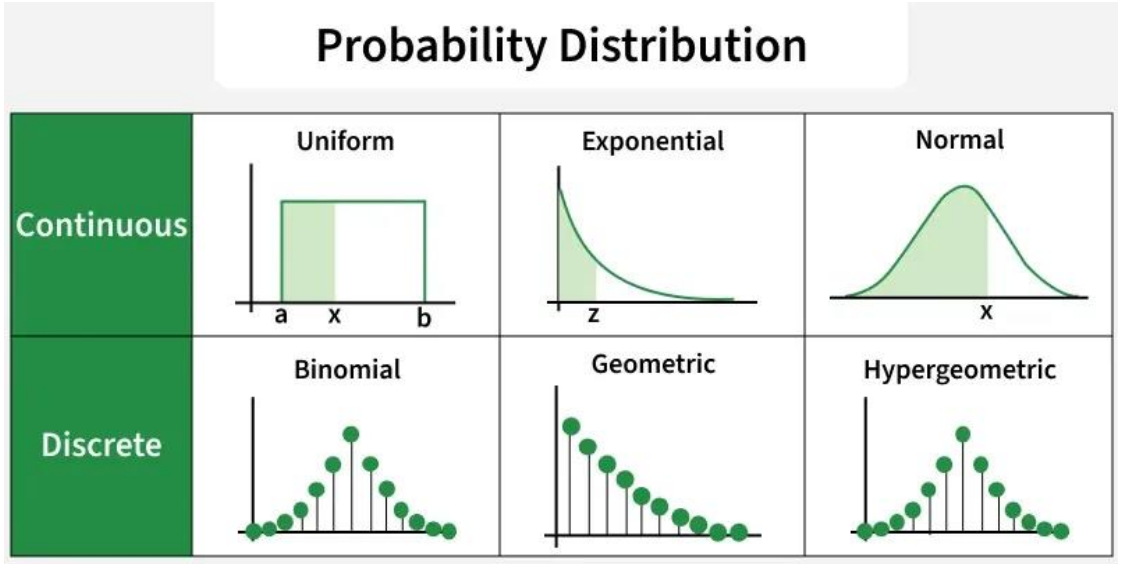
Selection bias (sampling bias)

- data used for training a machine learning model is not representative of the population it is intended to generalize to

Statistical definition – normal distribution







Four Ways to Draw a Representative Sample

Each method trades off precision, cost, and the need for a complete sampling frame

Simple random sampling



Every dot selected with equal probability — no pattern, no guarantees

Stratified sampling



Dots grouped by colour; proportional selection from each stratum

Cluster sampling



Geographic clusters selected as units — all dots inside move together

Quota sampling



Demographic targets set in advance; each group filled to quota

Types of data bias

Historical/temporal bias

- Bias present in the historical biases that have been encoded into the data. Not representative of the current context in terms of time

Implicit/Confirmation bias

- Amplifies pre-existing biases in the data or its creators, reinforcing patterns that align with its prior assumptions

Measurement bias

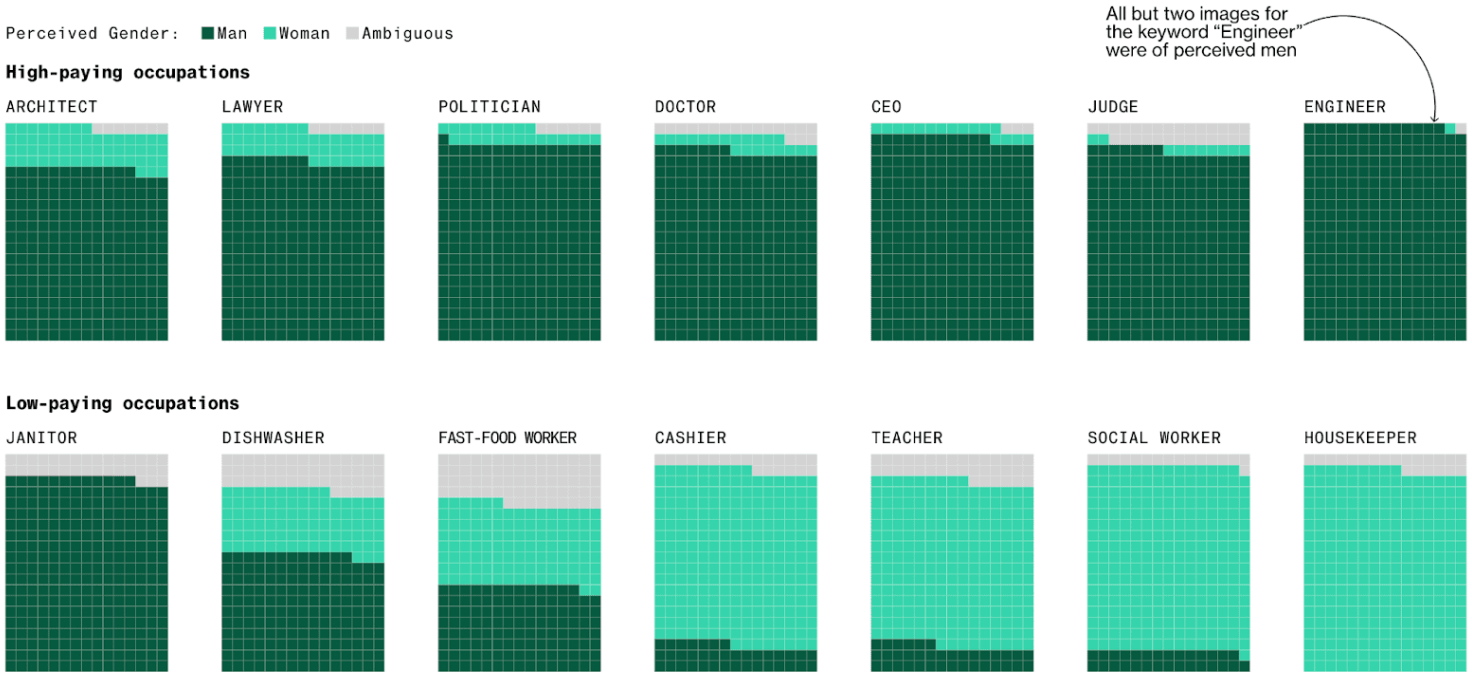
- Overrepresenting or underrepresenting certain populations or scenarios

Group Attribution Bias

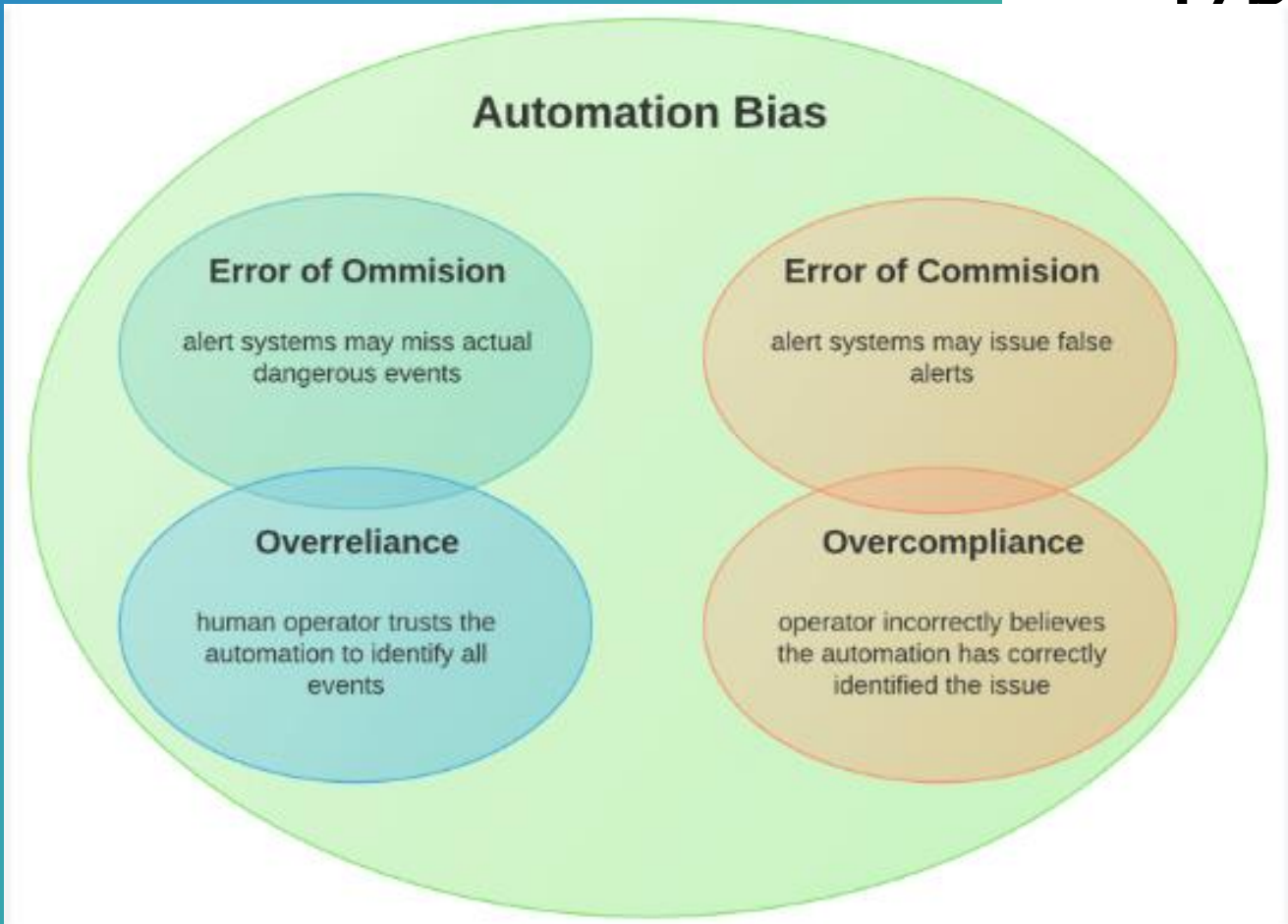
- AI system assumes that individuals within a group share the same characteristics or behaviors

Example

Example: Stable Diffusion and other generative AI models have been reported to exhibit socially biased behavior due to the quality of their training datasets. One study reported that the platform tends to underrepresent women in images of high-performing occupations and overrepresent darker-skinned people in images of low-wage workers and criminals. Part of the problem here seems to be the size of the training data. Generative AI models require massive amounts of training data and in order to achieve this data volume, the selection controls are often relaxed leading to poorer quality (i.e. more biased) input data.



Types of data bias



Automation bias

- the tendency to favor results generated by automated systems (generative AI data) over those generated by non-automated systems, irrespective of the relative quality of their outputs

Types of data bias

Generative Bias

- particularly generative models, when the content they create is unbalanced or misrepresented due to biased training data

Reporting Bias

- frequency or nature of events represented in a training dataset doesn't align with the real-world occurrence of those events

Automation Bias

- favor decisions made by automated systems over human judgment, even when the system's accuracy and/or reliability are questionable

Availability Bias

- we tend to put more work into understanding and recalling information in our working and short-term memory, as opposed to seeking information that may be harder to find

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7 STEPS TO MITIGATE AI BIAS

1. BIAS DETECTION

2. DEBIASING ALGORITHMS

3. DIVERSE TRAINING DATA

4. REGULAR AUDITS



5. TRANSPARENCY AND EXPLAINABILITY

6. ETHICAL AI PRACTICES

7. REINFORCEMENT LEARNING WITH RLHF

Strategies for reducing data bias



Recognizing and acknowledging data bias at design level



Investing in dataset diversity and data collection quality assurances.



Performing regular algorithmic auditing to identify and rectify bias.

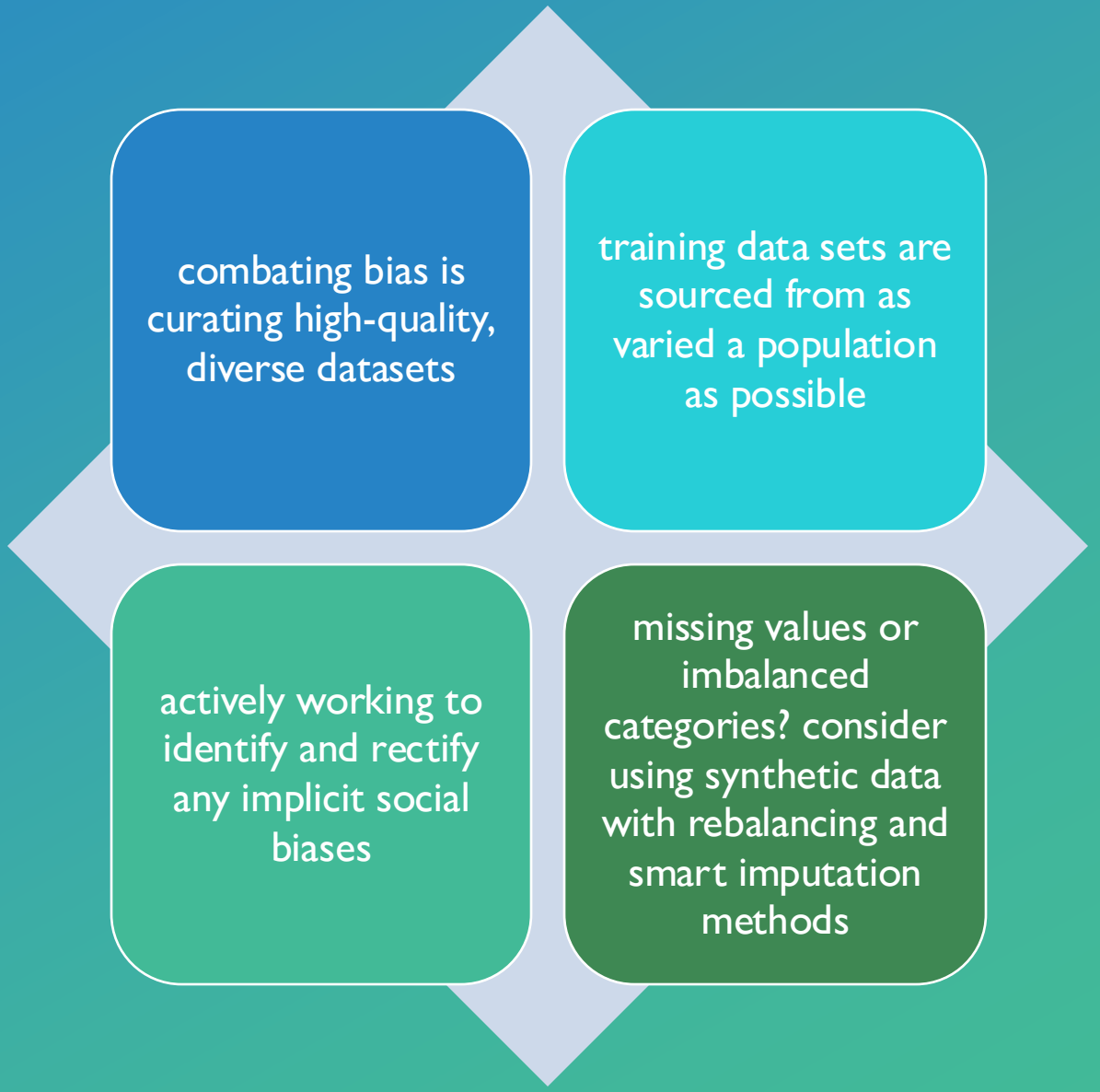


Including humans in the loop for supervision.



Investing in model explainability and transparency.

Diverse dataset curation



Tools for data bias and fairness

Check for overfitting, underfitting, outliers, recall, exclusion, measurement, association, social, historical biases, etc.

- AI Fairness 360 (AIF360) - A comprehensive toolkit by IBM for detecting and mitigating bias in machine learning models.
- Fairlearn - A Python library to assess and improve fairness in machine learning models.
- What-If Tool - An interactive tool by Google's PAIR team for investigating machine learning models and their fairness.
- FAT Forensics - A toolkit for assessing fairness, accountability, and transparency in AI systems.
- Themis-ML - A library for testing discrimination in machine learning models.
- Fairness Indicators - A suite of tools for evaluating the fairness of machine learning models in TensorFlow.
- Equality of Opportunity in Machine Learning - A toolkit for achieving fairness in predictive algorithms.
- The OpenAI Fairness Gym - A set of environments for studying the potential long-term impacts of AI algorithms on fairness.
- AI Explainability 360 (AIX360) - A toolkit by IBM for building explainable AI models and applications.
- Fair Accountability Design (FAccT) - Resources from the ACM Conference on Fairness, Accountability, and Transparency.

Video

- https://www.coursera.org/learn/global-mooc-on-the-ethics-of-ai/lecture/QKCqn/video-3-checking-ai-outcomes-by-demographic-group

Examples

- https://fat-forensics.org/sphinx_gallery_auto/accountability/xmpl_accountability_data_measure.html#sphx-glr-sphinx-gallery-auto-accountability-xmpl-accountability-data-measure-py
- https://aif360.readthedocs.io/en/latest/Getting%20Started.html
- https://github.com/Trusted-AI/AIF360/blob/main/examples/tutorial_credit_scoring.ipynb

Resources

- Case studies and tools: https://www.crescendo.ai/blog/ai-bias-examples-mitigation-guide#:~:text=Example%2014%3A%20Amazon's%20AI%2Dbiased,graduates%20of%20all%2Dwomen's%20colleges.